

EXAM 7 – SPRING 2013

6. (2.75 points)

Given the following information regarding a book of retrospectively rated policies:

	<u>Retrospective Adjustment</u>		
	<u>First</u>	<u>Second</u>	<u>Third</u>
Loss Age	18	30	42
Premium Age	27	39	51
Loss Capping Ratio	80%	50%	70%
LDF to Ultimate	1.800	1.150	1.050

- Tax multiplier: 1.04
- Loss conversion factor: 1.15
- Total basic premium: \$12,456,000
- Total standard premium: \$54,309,000
- Expected loss ratio to standard premium: 75%

a. (1 point)

Calculate the PDLD ratio as of the first retrospective adjustment.

b. (0.5 point)

Calculate the PDLD ratio as of the second retrospective adjustment.

c. (0.5 point)

Calculate the PDLD ratio as of the third retrospective adjustment.

d. (0.75 point)

Based on the results of parts a., b. and c. above, discuss the reasonableness of the loss capping ratios shown above.

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Question 6 Sample Answer

Solution 1

$$\begin{aligned} \text{a) } \text{PDLD}_{18} &= \frac{P_{18}}{L_{18}} = \frac{(BP + CL_{18} \times LCF) \times TM}{L_{18}} \\ &= \frac{12,456,000 \times 1.04 + 18,103,000 \times 1.15 \times 1.04}{22,628,750} = 1.53 \end{aligned}$$

$$E(L) = 75\% \times 54,309,000 = 40,731,750$$

$$\text{Loss}_{18} = 40,731,750 \times 1/1.8 = 22,628,750$$

$$CL_{18} = 22,628,750 \times 80\% = 18,103,000$$

$$\begin{aligned} \text{b) } \text{PDLD}_2 &= \left(\frac{CL_{30} - CL_{18}}{L_{30} - L_{18}} \right) \times LCF \times TM \\ &= \left(\frac{17,709,457 - 18,103,000}{35,418,913 - 22,628,750} \right) \times 1.15 \times 1.04 = -0.0368 \end{aligned}$$

$$CL_{30} = E(L)/1.15 \times 50\% = 17,709,457$$

$$L_{30} = E(L)/1.15 = 35,418,913$$

$$\text{c) } CL_{42} = E(L)/1.05 \times 0.7 = 27,154,500$$

$$L_{42} = E(L)/1.05 = 38,792,143$$

$$\begin{aligned} \text{PDLD} &= \left(\frac{CL_{42} - CL_{30}}{L_{42} - L_{30}} \right) \times LCF \times TM \\ &= \frac{27,154,500 - 17,709,457}{38,792,143 - 35,418,913} \times 1.15 \times 1.04 = 3.35 \end{aligned}$$

- d) Loss capping ratio should be decreasing ideally, however, in the question, 70% is higher than 50%. So, it's not reasonable.

Because as losses matures, more policies will hit the max premium and per occurrence limit and more losses will develop outside the caps. So capping ratios should decrease as losses mature.

Solution 2

$$\begin{aligned}
 \text{a) } \text{PDLD}_1 &= \frac{BP}{SP \times ELR \times \%L_1} \times TM + \frac{CL_1}{L_1} \times LCF \times TM \\
 &= \frac{12,456,000}{54,309,000 \times 75\% \times (1/1.8)} \times 1.04 + 0.8 \times 1.15 \times 1.04 \\
 &= 1.529268 \approx 1.53
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \text{PDLD}_2 &= \left(\frac{CL_2 - CL_1}{L_2 - L_1} \right) \times LCF \times TM \text{ (Assuming loss capping ratio given is incremental loss capping ratio)} \\
 &= 0.5 \times 1.15 \times 1.04 \\
 &= 0.598
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \text{PDLD}_3 &= \left(\frac{CL_3 - CL_2}{L_3 - L_2} \right) \times LCF \times TM \text{ (Assuming loss capping ratio given is incremental loss capping ratio)} \\
 &= 0.7 \times 1.15 \times 1.04 \\
 &= 0.8372
 \end{aligned}$$

- d) Typically we expect premium responsiveness to decline with increasing maturity, since as more time passes and more losses emerge, we expect more losses to be capped by the per-occurrence limit or the maximum premium, and only capped losses contribute to additional premium. Based on a-c above, we see that premium responsiveness at the 3rd adjustment is higher than the 2nd, which is contrary to the explanation above. This is

because the loss capping ratio is higher at the 3rd adjustment, suggesting fewer losses are being capped, which is not a reasonable assumption.

Examiner Comment

Part A, this is a relatively straightforward application of the formula for PDL ratio as provided within Teng and Perkins which calculates the Initial Premium as of the first adjustment as compared to the expected losses to have emerged as of that time. Common errors include arithmetic errors and errors in memorizing formulas.

Part B, the graders were expecting the candidates to understand the difference between cumulative and incremental activity. The true retrospective premium adjustments are based on the incremental activity from the prior adjustment to the subsequent adjustment. The loss capping ratios are cumulative unless they are labeled as incremental loss capping ratio. And candidates are expected to know how to use cumulative loss capping ratios to calculate the incremental ones. Common errors include confusing cumulative loss capping ratios to be incremental ones, forgetting TM and LCF, and arithmetic errors. For candidates who are using loss capping ratios as incremental ones, they must state their assumptions clearly; otherwise they will need to treat loss capping ratios as cumulative.

Part C, please see part B.

Part D, the candidates are expected to understand that the loss capping ratios given in the question were not reasonable in normal circumstance and why. The candidates are also expected to answer the question by using the data given in the question. Common errors will be candidates just giving generic comments/conclusions without tying them back to the situation that was given in the question.